STACKS, DISCRIMINANTS: LEMMA-NOETHER-PAIRING-COMPATIBILITIES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1385-1412 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 1414-1425, the complete author proof (10 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 44 of queue_discriminant.json (pb=479, stmt 1385-1412, proof 1414-1425).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0BVQ: sources/stacks-discriminant/pinned-original-source.tex, source lines 1367–1369. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVM: sources/stacks-discriminant/pinned-original-source.tex, source lines 1311–1320. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVL: sources/stacks-discriminant/pinned-original-source.tex, source lines 1295–1303. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVP: sources/stacks-discriminant/pinned-original-source.tex, source lines 1362–1369. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $A \to B$ be a quasi-finite homomorphism of Noetherian rings.*

Date: stacks-project:discriminant.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *If $A \to A'$ is a flat map of Noetherian rings, then*

$$\begin{array}{ccc} \omega_{B/A} \times J & \longrightarrow & B \\ \downarrow & & \downarrow \\ \omega_{B'/A'} \times J' & \longrightarrow & B' \end{array}$$

is commutative where notation as in Lemma 0BVM and horizontal arrows are given by (0BVQ).

(2) *If $B = B_1 \times B_2$, then*

$$\begin{array}{ccc} \omega_{B/A} \times J & \longrightarrow & B \\ \downarrow & & \downarrow \\ \omega_{B_i/A} \times J_i & \longrightarrow & B_i \end{array}$$

is commutative for $i = 1, 2$ where notation as in Lemma 0BVL and horizontal arrows are given by (0BVQ).

Proof. Because of the construction of the pairing in Remark 0BVP both (1) and (2) reduce to the case where $A \to B$ is finite. Then (1) follows from the fact that the contraction map $\mathrm{Hom}_A(M, A) \otimes_A M \otimes_A M \to M$, $\lambda \otimes m \otimes m' \mapsto \lambda(m)m'$ commuted with base change. To see (2) use that $J = J_1 \times J_2$ is contained in the summands $B_1 \otimes_A B_1$ and $B_2 \otimes_A B_2$ of $B \otimes_A B$. □